File

Name: P3-K02-helical
Description: KISSsoft example
Changed by: AAA on: 30.10.2026 at: 19:35:09

Micropitting according to ISO/TS 6336-22

1 Overview

Lubricant load according to FVA Info sheet 54/7 10, Mobilgear 600 XP 150
Reference data FZG-C Test:

Torque (Nm)	[T1Ref]	265.100
Line load at contact point A (N/mm)	[FbbRef,A]	236.300
Oil temperature (°C)	[θOilRef]	90.000
Tooth mass temperature (°C)	[θMRef]	128.037
Contact temperature (°C)	[θBRef,A]	253.806
Lubrication gap thickness (µm)	[hRef,A]	0.043
Specific film thickness in test	[λGFT]	0.085

1.1 Calculated for gear pair 1-2

Calculation at point (0:A, 1:AB, 2:B, 3:C, 4:D, 5:DE, 6:E, -1:No Point)			1	
Diameter (mm)	[dy]	40.877		162.962
Relative radius of curvature (mm)	[ρred]		5.162	
Load sharing factor	[XY]		0.658	
X_Y interpolated between the values $X_Y(\varepsilon_b=0.8)$ and $X_Y(\varepsilon_b=1.2)$ according to ISO/TC60/WG6.				
Contact stress (N/mm²)	[pH]		689.325	
Contact stress (N/mm²)	[pdyn]		947.689	
Minimal specific film thickness	[λGFY]		0.061	(h_Y=0.183 µm)
Safety against micropitting	[Sλ(B)]		0.511	

2 Calculation with nominal load

2.1 Calculated for gear pair 1-2

Equivalent Young's modulus (N/mm²)	[Er]	226373.6
Roughness average value (µm)	[Ra]	3.000
Elasticity coefficient (√N/mm)	[ZE]	189.812
Lubricant factor	[XL]	1.000
Lubrication coefficient for lubrication type	[XS]	1.000
Tip relief factor	[XCa]	1.817
Coefficient of friction	[µm]	0.106
Tooth mass temperature (°C)	[θM]	78.410
Dynamic viscosity (Ns/m²)	[ηtM]	23.768
Pressure-viscosity coefficient (m²/N)	[α38]	0.0205 $\cdot 10^{-6}$
Pressure-viscosity coefficient at mass temperature (m²/N)	[αθM]	0.0166 $\cdot 10^{-6}$

	A	AB	B	C	D	DE	E
g (mm)	0.000	1.298	2.595	2.973	4.547	5.845	7.142
Γ	-0.388	-0.218	-0.049	0.000	0.205	0.375	0.544
d_1 (mm)	40.193	40.877	41.712	41.982	43.232	44.403	45.692

d_2 (mm)	163.973	162.962	161.986	161.708	160.585	159.699	158.852
$\rho_{t,1}$ (mm)	4.695	5.992	7.290	7.668	9.242	10.540	11.837
$\rho_{t,2}$ (mm)	32.508	31.210	29.912	29.535	27.960	26.663	25.365
ρ_n (mm)	4.213	5.162	6.019	6.251	7.133	7.757	8.288
X	0.000	0.658	0.874	0.874	0.874	0.658	0.000
p_H (N/mm²)	0.000	689.325	736.036	722.255	676.134	562.349	0.000
p_{dyn} (N/mm²)	0.000	947.689	1011.908	992.961	929.553	773.121	0.000
$v_{r,1}$ (m/s)	1.450	1.851	2.252	2.369	2.855	3.256	3.657
$v_{r,2}$ (m/s)	2.607	2.503	2.399	2.369	2.242	2.138	2.034
v_g (m/s)	-1.157	-0.652	-0.147	0.000	0.613	1.118	1.622
$v_{\bar{\gamma}}$ (m/s)	4.057	4.354	4.651	4.737	5.098	5.394	5.691
θ_{fl} (Δ°C)	0.000	18.900	4.898	0.000	18.727	26.363	0.000
θ_B (°C)	78.410	97.310	83.308	78.410	97.137	104.773	78.410
G_M	3757.807	3757.807	3757.807	3757.807	3757.807	3757.807	3757.807
U	$50.564 \cdot 10^{-12}$	$44.280 \cdot 10^{-12}$	$40.566 \cdot 10^{-12}$	$39.787 \cdot 10^{-12}$	$37.518 \cdot 10^{-12}$	$36.509 \cdot 10^{-12}$	$36.049 \cdot 10^{-12}$
W	$0.000 \cdot 10^{-4}$	$1.101 \cdot 10^{-4}$	$1.255 \cdot 10^{-4}$	$1.209 \cdot 10^{-4}$	$1.059 \cdot 10^{-4}$	$0.733 \cdot 10^{-4}$	$0.000 \cdot 10^{-4}$
$\alpha_{\theta B}$ (m²/N)	$0.017 \cdot 10^{-6}$	$0.015 \cdot 10^{-6}$	$0.016 \cdot 10^{-6}$	$0.017 \cdot 10^{-6}$	$0.015 \cdot 10^{-6}$	$0.014 \cdot 10^{-6}$	$0.017 \cdot 10^{-6}$
S_{GF}	1.000	0.502	0.826	1.000	0.505	0.395	1.000
h (µm)	23.2448	0.1832	0.2204	0.2367	0.2269	0.2405	24.4047
λ_{GF}	7.748	0.061	0.073	0.079	0.076	0.080	8.135
S_λ	64.835	0.511	0.615	0.660	0.633	0.671	68.071

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